

Hallucination Detection Report

AI Output Verification · Knowledge Graph Analysis

Generated: 2026-04-19 00:34
Triples: 27 · Verdicts: 5 · Inconsistencies: 6 ·
Violations: 22

EXECUTIVE SUMMARY

5 Correct		
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EXTRACTED TRIPLES

All verifiable triples extracted from the LLM response before canonicalization or checking.

Subject	Predicate	Object	Verifiable	Verification Type	Predicate Definition
580	HAS_RISK_TIER	deep subprime	Yes	ENTITY_ATTRIBUTE	Indicates the risk tier category assigned to the subject based on credit score or other factors.
590	HAS_RISK_TIER	deep subprime	Yes	ENTITY_ATTRIBUTE	Indicates the risk tier category assigned to the subject based on credit score or other factors.
619	HAS_RISK_TIER	deep subprime	Yes	ENTITY_ATTRIBUTE	Indicates the risk tier category assigned to the subject based on credit score or other factors.
580	HAS_RISK_TIER	subprime	Yes	ENTITY_ATTRIBUTE	Indicates the risk tier category assigned to the subject based on credit score or other factors.
300	HAS_RISK_TIER	subprime	Yes	ENTITY_ATTRIBUTE	Indicates the risk tier category assigned to the subject based on credit score or other factors.
579	HAS_RISK_TIER	subprime	Yes	ENTITY_ATTRIBUTE	Indicates the risk tier category assigned to the subject based on credit score or other factors.
450	HAS_RISK_TIER	subprime	Yes	ENTITY_ATTRIBUTE	Indicates the risk tier category assigned to the subject based on credit score or other factors.
300	HAS_RISK_TIER	deep subprime	Yes	ENTITY_ATTRIBUTE	Indicates the risk tier category assigned to the subject based on credit score or other factors.
620	HAS_RISK_TIER	subprime	Yes	ENTITY_ATTRIBUTE	Indicates the risk tier category assigned to the subject based on credit score or other factors.
659	HAS_RISK_TIER	subprime	Yes	ENTITY_ATTRIBUTE	Indicates the risk tier category assigned to the subject based on credit score or other factors.
630	HAS_RISK_TIER	subprime	Yes	ENTITY_ATTRIBUTE	Indicates the risk tier category assigned to the subject based on credit score or other factors.
620	HAS_RISK_TIER	near prime	Yes	ENTITY_ATTRIBUTE	Indicates the risk tier category assigned to the subject based on credit score or other factors.
660	HAS_RISK_TIER	subprime	Yes	ENTITY_ATTRIBUTE	Indicates the risk tier category assigned to the subject based on credit score or other factors.
739	HAS_RISK_TIER	subprime	Yes	ENTITY_ATTRIBUTE	Indicates the risk tier category assigned to the subject based on credit score or other factors.
670	HAS_RISK_TIER	subprime	Yes	ENTITY_ATTRIBUTE	Indicates the risk tier category assigned to the subject based on credit score or other factors.
660	HAS_RISK_TIER	prime	Yes	ENTITY_ATTRIBUTE	Indicates the risk tier category assigned to the subject based on credit score or other factors.
740	HAS_RISK_TIER	subprime	Yes	ENTITY_ATTRIBUTE	Indicates the risk tier category assigned to the subject based on credit score or other factors.
850	HAS_RISK_TIER	subprime	Yes	ENTITY_ATTRIBUTE	Indicates the risk tier category assigned to the subject based on credit score or other factors.

Subject	Predicate	Object	Verifiable	Verification Type	Predicate Definition
750	HAS_RISK_TIER	subprime	Yes	ENTITY_ATTRIBUTE	Indicates the risk tier category assigned to the subject based on credit score or other factors.
740	HAS_RISK_TIER	super prime	Yes	ENTITY_ATTRIBUTE	Indicates the risk tier category assigned to the subject based on credit score or other factors.
900	HAS_RISK_TIER	subprime	Yes	ENTITY_ATTRIBUTE	Indicates the risk tier category assigned to the subject based on credit score or other factors.
200	HAS_RISK_TIER	super prime	Yes	ENTITY_ATTRIBUTE	Indicates the risk tier category assigned to the subject based on credit score or other factors.
300	HAS_INTEREST_RATE	-2.5	Yes	ENTITY_ATTRIBUTE	Indicates the interest rate applied to the subject.
850	HAS_INTEREST_RATE	40	Yes	ENTITY_ATTRIBUTE	Indicates the interest rate applied to the subject.
900	HAS_INTEREST_RATE	20	Yes	ENTITY_ATTRIBUTE	Indicates the interest rate applied to the subject.
200	HAS_INTEREST_RATE	20	Yes	ENTITY_ATTRIBUTE	Indicates the interest rate applied to the subject.
300	HAS_INTEREST_RATE	22.94	Yes	ENTITY_ATTRIBUTE	Indicates the interest rate applied to the subject.

CANONICALIZED TRIPLES

Predicates normalized against the database schema. Shows original predicate, canonical match, and match confidence.

Subject	Original Predicate	Object	Canonical Predicate	Confidence
580	HAS_RISK_TIER	deep subprime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH
590	HAS_RISK_TIER	deep subprime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH
619	HAS_RISK_TIER	deep subprime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH
580	HAS_RISK_TIER	subprime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH
300	HAS_RISK_TIER	subprime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH
579	HAS_RISK_TIER	subprime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH
450	HAS_RISK_TIER	subprime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH

Subject	Original Predicate	Object	Canonical Predicate	Confidence
300	HAS_RISK_TIER	deep subprime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH
620	HAS_RISK_TIER	subprime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH
659	HAS_RISK_TIER	subprime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH
630	HAS_RISK_TIER	subprime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH
620	HAS_RISK_TIER	near prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH
660	HAS_RISK_TIER	subprime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH
739	HAS_RISK_TIER	subprime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH
670	HAS_RISK_TIER	subprime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH
660	HAS_RISK_TIER	prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH
740	HAS_RISK_TIER	subprime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH
850	HAS_RISK_TIER	subprime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH
750	HAS_RISK_TIER	subprime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH
740	HAS_RISK_TIER	super prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH
900	HAS_RISK_TIER	subprime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH
200	HAS_RISK_TIER	super prime	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	HIGH
300	HAS_INTEREST_RATE	-2.5	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
850	HAS_INTEREST_RATE	40	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
900	HAS_INTEREST_RATE	20	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH
200	HAS_INTEREST_RATE	20	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH

Subject	Original Predicate	Object	Canonical Predicate	Confidence
300	HAS_INTEREST_RATE	22.94	INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE	HIGH

LOGICAL INCONSISTENCIES

Contradictions detected between triples within the LLM response, independent of the database.

Type	Severity	Explanation	Triples Involved
DIRECT_CONFLICT	HIGH	Subject '580' is classified as both 'deep subprime' and 'subprime' for the same risk tier, which is contradictory.	580 → RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL → deep subprime 580 → RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL → subprime
DIRECT_CONFLICT	HIGH	Subject '300' is classified as both 'subprime' and 'deep subprime' for the same risk tier, which is contradictory.	300 → RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL → subprime 300 → RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL → deep subprime
DIRECT_CONFLICT	HIGH	Subject '620' is classified as both 'subprime' and 'near prime' for the same risk tier, which is contradictory.	620 → RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL → subprime 620 → RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL → near prime
DIRECT_CONFLICT	HIGH	Subject '660' is classified as both 'subprime' and 'prime' for the same risk tier, which is contradictory.	660 → RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL → subprime 660 → RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL → prime
DIRECT_CONFLICT	HIGH	Subject '740' is classified as both 'subprime' and 'super prime' for the same risk tier, which is contradictory.	740 → RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL → subprime 740 → RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL → super prime
DIRECT_CONFLICT	HIGH	Subject '300' has two different annual borrowing interest rates, '-2.5' and '22.94', which is contradictory.	300 → INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE → -2.5 300 → INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE → 22.94

SYMBOLIC RULE VIOLATIONS

Violations caught by explicit domain constraints before database lookup. Violated triples were removed from the evidence retrieval step.

c_tier_580_619 • • MEDIUM

580 → RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL → deep subprime

Credit score 580 is in 580-619 range, but risk tier is 'deep subprime' instead of required 'subprime'.

c_tier_580_619 • • MEDIUM

590 → RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL → deep subprime

Credit score 590 is in 580-619 range, but risk tier is 'deep subprime' instead of required 'subprime'.

c_tier_580_619 · · MEDIUM

619 → RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL → deep subprime

Credit score 619 is in 580-619 range, but risk tier is 'deep subprime' instead of required 'subprime'.

c_tier_300_579 · · MEDIUM

300 → RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL → subprime

Credit score 300 is in 300-579 range, but risk tier is 'subprime' instead of required 'deep subprime'.

c_tier_300_579 · · MEDIUM

579 → RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL → subprime

Credit score 579 is in 300-579 range, but risk tier is 'subprime' instead of required 'deep subprime'.

c_tier_300_579 · · MEDIUM

450 → RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL → subprime

Credit score 450 is in 300-579 range, but risk tier is 'subprime' instead of required 'deep subprime'.

c_tier_620_659 · · MEDIUM

620 → RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL → subprime

Credit score 620 is in 620-659 range, but risk tier is 'subprime' instead of required 'near prime'.

c_tier_620_659 · · MEDIUM

659 → RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL → subprime

Credit score 659 is in 620-659 range, but risk tier is 'subprime' instead of required 'near prime'.

c_tier_620_659 · · MEDIUM

630 → RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL → subprime

Credit score 630 is in 620-659 but risk tier 'subprime' does not match required 'near prime'.

c_tier_660_739 · · MEDIUM

660 → RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL → subprime

Credit score 660 is in 660-739 but risk tier 'subprime' does not match required 'prime'.

c_tier_660_739 · · MEDIUM

739 → RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL → subprime

Credit score 739 is in 660-739 but risk tier 'subprime' does not match required 'prime'.

c_tier_660_739 · · MEDIUM

670 → RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL → subprime

Credit score 670 is in 660-739 but risk tier 'subprime' does not match required 'prime'.

c_tier_740_850 · · MEDIUM

740 → RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL → subprime

Credit score 740 is in 740-850 but risk tier 'subprime' does not match required 'super prime'.

c_tier_740_850 · · MEDIUM

850 → RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL → subprime

Credit score 850 is in 740-850 but risk tier 'subprime' does not match required 'super prime'.

c_tier_740_850 · · MEDIUM

750 → RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL → subprime

Credit score 750 is in 740-850 but risk tier 'subprime' does not match required 'super prime'.

c_risk_outside_range · · MEDIUM

900 → RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL → subprime

Credit score 900 is above maximum allowed (850), which always violates the range constraint.

c_risk_outside_range · · MEDIUM

200 → RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL → super prime

Credit score 200 is below minimum allowed (300), which always violates the range constraint.

c_interest_rate_range · · MEDIUM

300 → INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE → -2.5

Interest rate -2.5 is less than minimum allowed (0); this always violates the constraint.

c_interest_rate_range · · MEDIUM

850 → INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE → 40

Interest rate 40 is greater than maximum allowed (30); this always violates the constraint.

c_interest_rate_range · · MEDIUM

900 → INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE → 20

Credit score 900 is above maximum allowed (850); out-of-range subjects always violate the constraint.

c_interest_rate_range · · MEDIUM

200 → INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE → 20

Credit score 200 is below minimum allowed (300); out-of-range subjects always violate the constraint.

· · MEDIUM

300 → INTEREST_RATE_CHARGED_AS_ANNUAL_BORROWING_COST_PERCENTAGE → 22.94

Credit score 300 is within 300-850 and interest rate 22.94 is within range (0-30%), so the triple passes.

DETECTED HALLUCINATIONS

Subject	Predicate	Object	Verdict	Match	Violated	Conf.	Explanation
580	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	subprime	Correct	STRICT	—	1.00	Subject, predicate, and object all exactly match the DB candidate. Evidence: subject: 580, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: subprime
300	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	deep subprime	Correct	STRICT	—	1.00	All components match perfectly with DB record. Evidence: subject: 300, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: deep subprime
620	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	near prime	Correct	STRICT	—	1.00	Subject, predicate, and object strictly match the DB candidate. Evidence: subject: 620, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: near prime
660	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	prime	Correct	STRICT	—	1.00	All triple elements are identical to DB entry. Evidence: subject: 660, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: prime
740	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	super prime	Correct	STRICT	—	1.00	Subject, predicate, and object match exactly with DB. Evidence: subject: 740, predicate: RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL, object: super prime

TRIPLE AUDIT TRAIL — ORIGINAL → CANONICAL → VERDICT

Full pipeline trace for every triple. UNVERIFIABLE* means the triple had no verdict — e.g. dropped due to output truncation.

Orig. Subject	Orig. Predicate	Orig. Object	Confidence	Canon. Predicate	Canon. Object	Verdict
580	HAS_RISK_TIER	subprime	HIGH	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	subprime	CORRECT
300	HAS_RISK_TIER	deep subprime	HIGH	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	deep subprime	CORRECT
620	HAS_RISK_TIER	near prime	HIGH	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	near prime	CORRECT
660	HAS_RISK_TIER	prime	HIGH	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	prime	CORRECT
740	HAS_RISK_TIER	super prime	HIGH	RISK_TIER_CLASSIFIED_AS_BORROWER_DANGER_LEVEL	super prime	CORRECT

VERDICT LEGEND

Verdict	Description
Correct	All three components (subject, predicate, object) verified against DB.
Rule Violation	One or more components contradict DB evidence.
Multi-Rule Violation	Violation spans multiple DB triples simultaneously.
Unverifiable	No usable DB evidence found for any component.
Predicate Mismatch	Subject real, but stated predicate is not a valid relation for it.
Fabricated Fact	Triple asserts a relationship that does not exist anywhere in DB.